

Ali Zamani

Senior Machine Learning Engineer — Agentic AI

LLMs | Multi-Agent Systems | RAG | MLOps | Production ML

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Summary

Senior Machine Learning Engineer with 5+ years in production ML, the last 3 focused on LLMs, RAG, and multi-agent systems on GCP (Vertex AI) across retail, travel, and healthcare. Built a RAG retrieval pipeline at 97% recall serving 10K daily queries; shipped a GenAI feature that lifted conversion by 30%, validated via A/B test.

Experience

Senior Machine Learning Engineer

Jun. 2025 – Present

Loblaw Digital, Toronto, ON

- Architected a **planner-based multi-agent AI platform** (specialized domain agents, multi-layer guardrails, hierarchical long/short-term/episodic memory) supporting **7 domain agents** in production.
- Led build-vs-buy decision and built a production **hybrid-search Agentic RAG** platform on **Vertex AI, SPLADE, Qdrant, Cloud Run, and Airflow**; cut projected infra costs by **\$75K–\$85K/yr** at **97% recall** and **78% precision**.
- Built a reusable **Agent Tool Framework (LangGraph, MCP)** for standardized tool integration, adopted by **2 teams** across **4 AI initiatives**; cut new-tool integration time from **10 hours to 30 minutes**.
- Built an **XGBoost** query-routing service with Search and Backend teams covering **3 categories**, reducing misrouted queries by **60%**.

Machine Learning Engineer

Apr. 2023 – Jun. 2025

Priceline (Booking Holdings), Toronto, ON

- Built a reusable **ML platform** on **Vertex AI** (preprocessing, training, feature engineering, versioning, backfilling, online/offline deployment via **Airflow**); cut model deployment time from **30 days to 1 day** across **5 use cases**.
- Architected a **GenAI framework** on GCP (CI/CD, production inference, **LLM-as-a-Judge** eval, rate limiting); now powers **5 production LLM workflows**.
- Shipped production **hotel review summarization** (sentiment analysis + LLM summarization) driving **30% conversion uplift**, validated by A/B test.

Data Scientist

Nov. 2022 – Apr. 2023

AltaML, Toronto, ON

- Designed and deployed a **semantic search engine** and **document Q&A pipeline (LangChain, OpenAI GPT, FAISS/Pinecone)** for a major banking client, reducing reliance on manual support.
- Built a modular **evaluation framework** for LLM Q&A (retrieval quality, groundedness, answer relevance).
- Implemented **LightGBM** for Mean Grain Size estimation from core images at **94% accuracy**, contributing to an estimated **\$10M+** in client value.

Data Scientist and Chatbot Developer (MIRA Chatbot -)

Jan. 2021 – Sept. 2022

Department of Computing Science, University of Alberta and Amii, Edmonton, AB

- Built and deployed the full-stack **MIRA chatbot** (mymira.ca) end-to-end, from NLP pipeline to production front-end.
- Evaluated **RNN** architectures for intent detection and entity extraction, reaching **97%** and **83% F1** respectively.

Education

M.Sc. in Computer Science

University of Alberta – Thesis: MIRA Chatbot, advised by Dr. Osmar R. Zaiane

Jan. 2021 – Aug. 2022

Edmonton, AB, Canada

M.Sc. in Computer Engineering

Amirkabir University of Technology – Thesis: Computer Network Optimization

Sep. 2017 – Sep. 2020

Tehran, Iran

B.Sc. in Electrical Engineering

Kashan University

Sep. 2013 – Sep. 2017

Isfahan, Iran

Technical Skills

Programming: Python (5+ years), C++, C, JavaScript

LLM / GenAI / Agents: LangChain, LangGraph, MCP, prompt engineering, fine-tuning, RLHF, agent orchestration, RAG, LLM-as-a-Judge, multimodal models

ML / Modeling: TensorFlow, PyTorch, Hugging Face, Scikit-learn, XGBoost, LightGBM, SPLADE, time series forecasting

Cloud / Infra: GCP (Vertex AI, BigQuery, Cloud Storage), Azure, Docker, Kubernetes, Airflow, Cloud Run

Data / Search: Qdrant, SQL, PostgreSQL, MySQL, BigQuery, Spark

Tools: Git, NGINX, Jira

Leadership: Mentored 2 junior engineers, led cross-functional design reviews across 3 teams

Selected Projects

plancontract

- Framework-agnostic **Pydantic schemas and validators** for multi-agent turn routing plans, standardizing which agent acts next in LLM orchestration pipelines.

Google Hackathon — 1st Place (2023) and 3rd Place (2022)

- Built a vector-embedding **neighborhood recommendation engine** and a **GPT-4 travel assistance chatbot** for conversational destination discovery.

ML Pipeline Template

- Reusable ML pipeline template accelerating model development and deployment across problem types.

Microsoft Hackathon

- **Burnout Detection System:** Azure ML pipeline with fine-tuned BERT for sentiment/stress detection in call-center conversations at 90% accuracy.

Selected Publications

Developing and Implementing a Mental Health Chatbot to Support Healthcare Workers [\[LINK\]](#)

2021, A.Zamani, M.Gharayat, J.Nobel, O.Zaiane, E.Stroulia – REMAP-D, Vancouver, BC

M.Sc. Thesis: MIRA Chatbot for Mental Health Support [\[LINK\]](#)

2022, A. Zamani, advised by Dr. Osmar R. Zaiane – University of Alberta

An Efficient Load Balancing Approach for Service Function Chain Mapping [\[LINK\]](#)

2021, Ali Zamani, Bahador Bakhshi, Saeed Sharifian